

An Android-Based Number Conversion System to Yorùbá Language Numeral Forms

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Abstract

The Yorùbá numeral system has various forms of usage and the different forms of this usage are already becoming a thing forgotten by most of the speakers of the language. To bring this awareness of this usage to these speakers, this study developed an android application that can translate a Hindu-Arabic number or English number text to either simple enumeration, adjectival, adverb of number, adverb of time, ordinal, or all forms. Data of various Yorùbá forms were collected from some Yorùbá textbooks. The system was designed using Unified Modelling Language (UML) tools, implemented using Android Studio, and evaluated using the Bilingual Evaluation Understudy (BLEU). The evaluation result shows that the application has a BLEU score of 0.94 for simple enumeration, 1.00 for ordinal, 1.00 for Adjectival, 0.91 for adverb of time, 0.90 for adverb of number, 0.93 for all, and 1.00 for numismatics.

1.0 Introduction

The *Yorùbá* language is one of the languages with the highest number of speakers in Nigeria. According to Webster [1], a number is a word or symbol that represents a specific amount or quantity. A numeral is a conventional symbol that represents a number. According to Murray [2], a number is a symbol, word, or figure representing a particular quantity and is used in counting and making calculations. The numeral system of any language can thus be regarded as a number, naming scheme within a particular language that specifies how numbers are spoken and written within such language. It is an abstract concept used in the representation of the quantity of objects and has been used for counting and measuring for more than 5,000 years by some cultures for communication in education, trading, and agriculture [3].

Elésèmoyò [4] stated that the counting system of the *Yorùbá* Language is an outstanding numeral system that attracts many linguists [5], [6], [7], [8], [9], [10], [11] and recent researchers like [12], [13], [14], [15], [16], [17]. The *Yorùbá* language has become a second language for most speakers and a reasonable population of the *Yorùbá* people do not speak the language [4]. The English language has become the only language for some. The English language has taken over as the medium of communication in schools, courtrooms, legislative houses, hospitals, and so on. The numeral system suffers more because it is considered difficult to learn due to its complex computation (i.e. usage of subtraction and over-counting) even though some local speakers of the language without formal education have no such

difficulty, especially for their daily business and trading [4]. The beauty of the counting system is the various forms of usage it has and the linguistic characteristics of each, but a number of users are not aware of the various forms and their usage. For this reason, those forms of usage are going extinct. Some researchers have ventured into digitising the number system, while some focused on majorly one of the counting systems usually the numismatic [16] or adjectival [14], [18] for others have scope of range from 1 to 1000 except for Akinadé and Odejobí [14] that has capacity beyond 1000. Because of the complexity and scarceness of resources for the numeral system, this study employed computational and linguistic processing to develop an android application for speakers of the language to learn the various forms of usage of the numeral system and covers a range beyond 1 million.

2. Literature Review

Akinadé and Odejobí [14] worked on the Computational modelling of *Yorùbá* numerals in a number-to-text conversion system. This work focused on the conversion of Hindu-Arabic number digits like 1234, 234, 568, and 5, which were translated to the corresponding *Yorùbá* text like *ẹgbẹ̀fà ó lé m̀ẹ̀rìnlélógbòn*, *igba ó lé m̀ẹ̀rìnlélógbòn*, *ẹ̀ẹ̀dẹ̀gbẹ̀ta ó lé m̀ẹ̀jídínlàádòrìn* and *márùn*. The work considered the complexities involved in big numbers to *Yorùbá* text which so many speakers of the language generate intuitively rather than have a uniform way of generation. There are infinite ways in which large numbers can be generated. Since *Yorùbá* numeral makes use of subtraction for over-counting, the work further considered the level of computation involved in generating the different translations of the

large numbers and also the depth of the parse tree generated for the translated number.

Elesemoyo and Odejobi [15] developed a system to translate English sentences with numbers in English text to the corresponding *Yorùbá* form. There are four number forms covered in the work and they are: adjectival form, adverb of time form, adverb of number form and all-form of the *Yorùbá* numeral system. The evaluation of the system developed produced BLEU scores of 0.946, 0.948, 0.902 and 0.534 for adjectival form sentences, adverb of-number form sentences, adverb of-time form sentences and all-form sentences respectively.

Jimoh *et al.* [16] developed an Android application to translate numbers to *Yorùbá* text. The scope of the work is the conversion of numbers from 1 to 1000 to the simple enumeration form of the counting system.

Elesemoyo and Odejobi [17] compared the *Yorùbá* counting system with the Roman, Chinese, Hindu-Arabic, Hausa, and Igbo counting systems. The work proves that the *Yorùbá* counting system is more complex than the other counting systems it was compared with. The reason given for this conclusion is that the *Yorùbá* counting system is a complex one that cascades three number bases (five, ten, and twenty). Furthermore, Elesemoyo and Odejobi [17] posited that the counting system uses subtraction extensively and does over-counting.

Agbeyangi *et al.* [18] worked on the development of an English to *Yorùbá* Text-to-Text Translation system. This work focused on the conversion of English number text to the corresponding *Yorùbá* text. The work in this study went further to consider the

translation of English number text in a phrasal context to the corresponding *Yorùbá* text.

The *Yorùbá* people use numbers for ordinal (positioning) and cardinal (counting) purposes. The cardinal purpose is further divided into six forms of usage [19], [20], [21].

2.1 Simple Enumeration Form

This is the first category of the number form. This is more like the simplest ways in which *Yorùbá* numbers occur. The other forms are derived from this form by way of adding prefixes. E.g. *èjì* under this form becomes *méjì* under the numeral adjectives form.

2.2 Numeral Adjectives Form

Numbers in this form qualify the noun following them usually an object or entity. The numeral adjective form is formed from the concept of taking the simple enumeration form. e.g. *mú èjì* which means “take two”. It is thereby shortened to *méjì* by eliminating the “u”. The letter “m” is added as a prefix to simple enumeration form of the numbers 1 to 10 or numbers preceded either by the addition of the *Yorùbá* equivalence of 1 to 4 or by the subtraction of the *Yorùbá* equivalence of 1 to 5.

2.3 Numismatics Form

Numbers in this form are used for trading and commerce. They are used to address the number of cowries which is the means of exchange in those days. An example of this is two cowries which means *owó èjì* which is then shortened as *eéjì*.

2.4 Adverb of number Form

Numbers in this form qualify the verb in the sentences. The number takes the adjectival form and duplicates it from one to nineteen, and then the first two letters are added to the case of 20, 30, 40, 60, 80, 100, 120, 140, 160, and 180. In the case of 50, 70, 90, 110, 130, 150, 170, and 190 the first two letters are duplicated and the consonant is changed to the letter “r” and then re-added to the number. The numbers are also reduplicated for 21-29, 31-39, 41-49, and so on.

2.5 Adverb of Time Form

Numbers in this form like adverb of number form qualify the noun in the sentence, but this form is with respect to the number of times the action was performed. The *ẹẹ* is prefixed to all the numerals but the multiples on ten take *ìgbà* before them.

2.6 All-Form

Numbers in this form are used to qualify all objects in a group. In this sense everyone, every object is taken into account. Examples of the case are “all five” and “the six” and here there are five entities in the first instance and six in the second. The formation of all forms is done by inserting the consonant starting the second syllable and the vowel ending the first syllable with the low tone changed to high. This is performed on the adjectival numbers starting with “m” excluding fifteen because the second syllable is a vowel and not a set of letters starting with a consonant. Other multiples of ten are preceded by the word *gbogbo*.

2.7 Ordinal Form

Numbers in this form are used for object positioning. Examples are First - *ìkínní*, Thirty-fifth - *ìkarùndínlógójì*.

3. Methodology

The methods described in the subsections of this section were used to achieve the aim of this study.

3.1. Data Collection

Data of various forms in which *Yorùbá* language use numbers were collected from [19], [20], and [21].

3.2 System Design

The conceptual model of the system was designed as shown in Figure 1. The first stage of the system working is the input of an English textual number or a number digit. If the input is a textual number, the system will first convert the number to digits using Algorithm 3.1 before proceeding to the next operation, which is the decomposition of the number according to the number decomposition algorithm in [14]. The next stage is to convert the decomposed numbers to their respective *Yorùbá* lexical equivalence after which linguistic rules will be applied based on the selected form to give the expected *Yorùbá* translation of the input number based on the chosen form. The flowchart is shown in Figure 2. It takes input text and the form to be translated to is chosen, if the input text is not a number digit, it will be converted to a number digit and thereafter it will be decomposed, converted to token equivalence, and then linguistic rules for the selected forms will be applied and the output will be displayed.

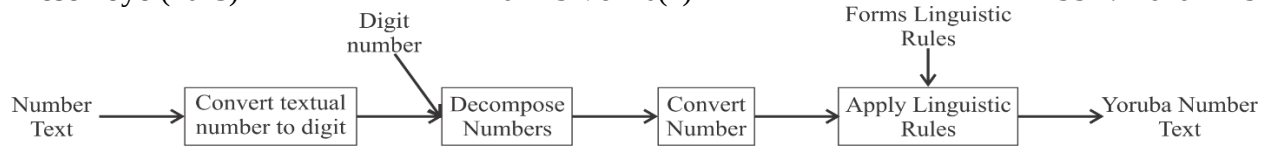


Figure 1: Number to Yorùbá number forms translation model

3.2.1. Textual Number to Digit

If the input is a textual number, the text is first converted to digits before it undergoes the conversion process. The textual number is tokenized, and the token “and” is removed if it is present in the list of tokens into which the number text is broken down. For example, the list of tokens [‘six’, ‘thousand’, ‘eighty’, ‘hundred’, ‘and’, ‘sixty’] becomes [6, 1000, 8, 100, 60] which is further reduced to [6000, 800, 60], these are then added together to form the digit 6860 which is the number equivalent of the text. This is shown in Algorithm 3.1.

3.2.2. Number Decomposition

Number decomposition involves the breaking down of number digits into smaller numbers. According to Akinadé and Odejóbi [14], the first process here is to generate a magnitude stack for the given number. The magnitude stack contains numbers that can be derived as multiples of multiples of Yorùbá numeral multiplicative bases (i.e. 1, 20, 200, 2000, 20000). The number decomposition process will generate 5 new numbers (d_0, d_1, d_2, d_3, d_4) for the given number.

So, we can have

$$\text{number} = 20000 * d_4 + 2000 * d_3 + 200 * d_2 + 20 * d_1 + d_0$$

such that d_0, d_1, d_2, d_3, d_4 can be expressed using just basic lexical items of the Yorùbá numeral system. The magnitude generator algorithm in [14] was used for the number breakdown. Any of the items in the stack whose value is zero will be removed from the stack. For example, the magnitude stack generated for 76,546 and 123 were $[d_4, d_3, d_2, d_1, d_0] = [3, 8, 2, 7, 6]$ and $[d_4, d_3, d_2, d_1, d_0] = [0, 0, 0, 6, 3]$.

3.2.3. Conversion of Numbers

The conversion of lexical items of the forms stack was achieved by the use of four dictionaries containing the basic lexical items (DIGIT), summative base twenty (D), multiplicative base (M), and the operational connectives (OPERATORS).

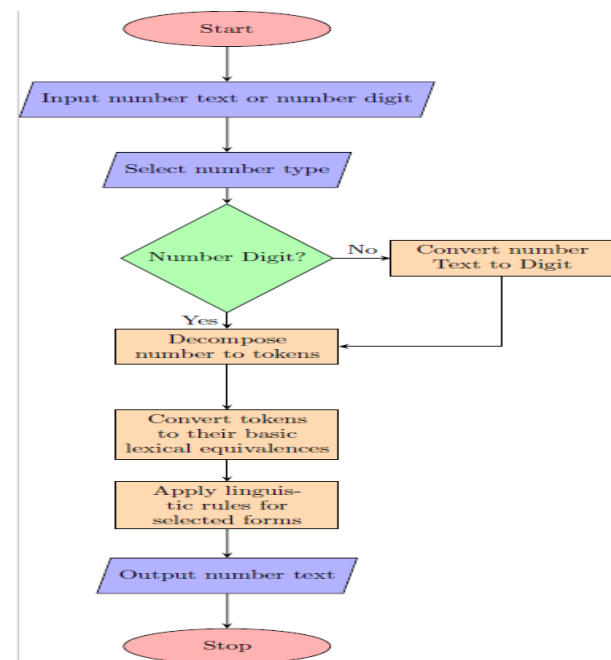


Figure 2: Flowchart

DIGIT = {1:òkan, 2:èjì, 3:èta, 4:èrin, 5:àrún, 6:èfà, 7:èje, 8:èjò, 9:èsán, 10:èwá, 20:ogún, 30:ogbòn, 200:igba, 300:òdúnrún, 400:irínwó, 20000:òké}

D = {D20:okòó, D40:òjì, D60:òtà, D80:òrìn }

M = {20*:ogún*, 200*:igba*, 2000*:egbàá, 20000*:òké}

OPERATORS = {+:lé ní, -:dín ní, ++: ó lé, -:ó dín, 10-:àádín, 100-:èédín, 1000-:èédín}

3.2.4. Linguistic Rules

Some linguistic rules were applied in achieving the translation of the number text to the respective forms. Conversion of 54 for example to simple enumeration form is as follow. The number is decomposed to [4, 50]. This is further processed [4 + 10 - (20 * 3)] which is four added to ten removed from trice twenty. Each of this are replaced with their lexical equivalence from the map data structures in subsection 3.2.3. Therefore, we have [èrin lé ní àádín ogún èta]. Lastly some morphophonological rules are linguistic manipulation such as:

- i. **vowel coalescence** : is a process where the vowel that ends a morpheme and the vowel that begins the next morpheme are fused together to form another vowel (e.g ún + è → o; ogún + èta → ogóta),
- ii. **Deletion**: is a process of shortening a phrase or word by deleting a segment (e.g lé ní + àádín → lélàádín, here í is deleted and n is converted to l),
- iii. **Vowel harmony**: based on the classification of standard Yorùbá vowel in [22] and [23] into advanced

tongue root (ATR) like e, i, o, u and non-ATR a, ẹ, ọ. After the vowel coalescence of ogún + èta → ogóta, we have two non-ATR (ó, a) and one ATR (o) vowels the ATR vowel is converted to (ọ) its non-ATR equivalence to harmonise the vowels thereby changing ogóta to ọgóta.

- iv. **Vowel assimilation**: this is a process where a vowel completely or partially becomes another [23]. E.g. igba èrin, the vowel a is deleted and i is assimilated to ẹ to form ẹgbèrin.

3.2.4.1 Simple Enumeration

This is the simplest form of the counting system. It is used for goods counting. The process in the beginning part of this subsection will produce the simple enumeration form of the Yorùbá number system as the form is the simplest form of the counting system.

3.2.4.2 Adjectival form

The adjectival form of the Yorùbá numeral takes the simple short form of the Yorùbá numeral in the form òkan, èjì, èta, etc. The ò in òkan is removed while all the numbers that are prefixed with èjì to èwá in the simple enumeration form are further prefixed by “m” and the vowel starting the number is changed from low tone to high tone and every other number of the multiples of 10 retains their name.

3.2.4.3 Numismatic

The ení in simple enumeration was replaced with oókan, while the vowel letter starting the simple enumeration form of numbers from two to ten, the first four numbers following a multiple of ten (e.g. from 11 to 14, 21 to 24,

31 to 34, and so on) and last four number ending a ten as from twenty (e.g. from 15 to 19, 25 to 29, 35 to 39 etc.) which takes a low tone (̀) with two of the same vowel letter (the first having mid-tone with no tone mark and the second having the high tone mark (´)). E.g. ẹ replaced with ẹẹ in the equivalence of three where ẹta becomes ẹẹta.

3.2.4.4 Ordinal

The ọkan of Adjectival is replaced with *ikíní* and the “m” that serves as the prefix for the adjectival is replaced with “ik”.

3.2.4.5 Adverb of Number form

The formation of the adverb of number form is done by duplicating numbers from two to ten, duplicating the first two letters of the multiples of twenty till 180, prefixing the numbers starting with *àád* and *ẹẹd* with *àr* and *ẹr* respectively.

Algorithm 3.1 English number text to number digit conversion

1. **Input:** numberSentence
2. **Output:** Number
3. **procedure**
tokenizeNumber(numberSentence)
4. words = numberSentence.split()
5. **return** ([SingleWordNumber(i)] for i
in words) if (i != “and”)
6. **end procedure**
7. **input:** numbertokens
8. **output:** tokens
9. **procedure** coalesceTokens(tokens)

10. result = []
 11. **for** token in tokens **do**
 12. **if** result and token == 100 and
result[-1] < 100 **then**
 13. result[-1]* = 100
 14. **else if** result and result[-1] < 1000
and token < 1000 **then**
 15. result[-1]+ = token
 16. **else**
 17. result.append(token)
 18. **return** result
 19. **end procedure**
 20. **Input:** tokens
 21. **Output:** Number
 22. **procedure** parse(tokens)
 23. Number = 0
 24. **while** tokens **do**
 25. **if** tokens.length == 1 or tokens[0] ≥
1000 **then**
 26. total+ = tokens.pop(0)
 27. **else**
 28. **assert** tokens[1] >= 1000
 29. total+ = tokens[0] * tokens[1]
 30. **del** tokens[:2]
 31. **end if**
 32. **end while**
 33. **end procedure**
-

3.2.4.6 Adverb of Time form

The formation of the adverb of time form is done by adding a prefix of *ẹẹ* to the adjectival form of numbers such as one and every other number that starts with an “m”. Every other number that is a multiple of 10 is preceded by the word *ìgbà*.

3.2.4.7 All form

The formation of all forms is done by inserting the consonant starting the second syllable and two of the vowel ending the first syllable with their high tones changed to low after the first syllable. This occurs for *Yorùbá* adjectival numbers starting with “m” excluding fifteen because the second syllable is a vowel and not a set of letters starting with a consonant. For example, *méjì* is the adjectival form equivalent for two (2), the consonant starting the second syllable is “j” while the vowel ending the first syllable is “é”. Inserting “j” and double “é” changed to low tone “èè” after the first syllable will give *méjèèjì* which is the all form for two (2). Other multiples of ten retains their simple enumeration form preceded by the word *gbogbo*. For example, twenty (20) is *gbogbo ogún*.

3.3 Implementation

The application was implemented using Android Studio software and Java programming language.

3.4 Evaluation

The system is evaluated using the Bilingual Evaluation Understudy (BLEU). This is done on character level since some of the samples

under considerations are lone words. The n-gram precision for BLEU is calculated as

$$P_n = \frac{\sum_{c \in C} \min(\text{count}(c, \text{candidate}), \text{count}(c, \text{reference}))}{\sum_{c \in C} \text{count}(c, \text{candidate})}$$

$$BLEU = BP \cdot \exp\left(\sum_{n=1}^N w_n \log P_n\right)$$

Where:

- P_n is the modified n-gram precision
- c is an n-gram (character sequence of length n)
- $\text{count}(c, \text{candidate})$ is the number of times n-gram c appears in the candidate
- $\text{count}(c, \text{reference})$ is the number of times n-gram c appears in the reference
- The minimum count ensures no overestimation
- w_n is weight for each n-gram precision
- BP is the brevity penalty

4. Results

Sample outputs of the System are shown in Figure 3. Figures 3a, 3b, 3c, and 3d show the adverb of number output of 123, adverb of time output of 46, all form output of 123 and ordinal output of 46 respectively. Sample wrong output is shown in Figure 4, the system produced *òkẹ ení* instead of *òkẹ kan*. Bilingual Evaluation Understudy (BLEU) was used to evaluate the performance of the

System. Ten random numbers were used for the evaluation of the system. The numbers 1, 5, 10, 40, 46, 54, 123, 534, 1000, and 20,000 were supplied for the seven different forms. The cumulative BLEU Scores for the forms

are shown in Table 1. The application developed has a minimum of 0.90 BLEU score which was achieved on the Adverb of Number and achieved a BLEU score of 1 for Ordinal, Adjectival, and Numismatics.

(a) Adverb of Number output for 123

(b) Adverb of Time output for 46

(c) All form output for 123

(d) Ordinal output for 46

Figure 3: Sample System Outputs

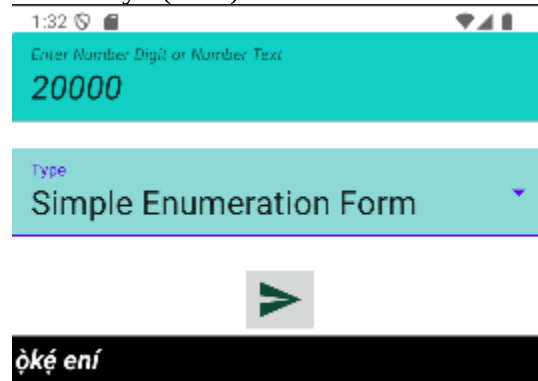


Figure 4: Sample incorrect output for 20000 to simple enumeration form

Table 1. BLEU Score for the forms

S/N	Form	Bleu Score
1	Simple Enumeration	0.94
2	Ordinal	1.00
3	Adjectival	1.00
4	Adverb of Number	0.90
5	Adverb of Time	0.91
6	All	0.93
7	Numismatics	1.00

5. Discussion

The result obtained from the system indicated the effectiveness of the methodology adopted. The high BLEU values for most numeral forms demonstrate that the system can effectively translate numbers from Hindu-Arabic numerals and English textual numbers to corresponding Yoruba numeral forms. The ideal BLEU score of 1.00 for Ordinal, Adjectival and Numismatics forms show that the system works well for these forms. The lowest BLEU score of 0.90 for

Adverb of Number form, while relatively lower, still shows a very great degree of precision in translation. The small discrepancies observed, such as the incorrect generation of "òkè ení" instead of "òkè kan" for 20,000. This error is due to the fact that one is translated under simple enumeration form as *ení*. But in the case of multiples of 20000, the *Yorùbá* number system uses an additional number (a multiplier) to explain the number multiplying 20000 to get the number under consideration. The multiplier number is represented in

adjectival form no matter the form under consideration. In the case of 20000, the multiplier is one (1) and it is translated as kan to produce òkẹ́ kan.

This study shows that the Yorùbá number text for these various forms can be computationally and linguistically generated by supplying the Hindu-Arabic equivalent of the number. This work covers 7 numbers form usages in the Yorùbá language and extends beyond 1 million compared to Akinade *et al.* [14], Agbeyangi *et al.* [18] which considers the adjectival form and Jimoh *et al.* [16] which covers the numismatic form and is limited to a range of 1 to 1000 of the number system.

6. Conclusion

This paper analysed the conversion of Hindu-Arabic Numeral or English number text into seven various forms: Simple Enumeration, Ordinal, Adjectival, Adverb of Number, Adverb of Time, All, and Numismatic. An android application was developed to accept the number to be converted either as a textual number or a digit number and the form to which the number is to be translated. The number is decomposed and saved into a magnitude stack, the magnitude stack tokens are converted using lexical replacements, and then the linguistic rule of the selected form is applied to the result of the lexical output to produce the required output.

These various number words can be useful in text normalisation for the Yorùbá language. It is necessary to apply the application as a teaching aid for the Yorùbá numeral system and evaluate its effectiveness in aiding the teaching of the numeral system both for

younger students and students who are learning Yorùbá as a second language.

Declarations

Ethics approval and consent to participate

Not Applicable

Consent for publication

Not Applicable

Availability of data and materials

The data used are available in the sources cited.

Competing interests

The author declare that they have no competing interests.

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Authors' contributions

The author is the sole author and the whole idea of the research was conceived and carried out by the author.

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References

- [1] Webster, M. (2011). *Merriam-Webster's Dictionary*, eleventh ed., Springfield, Massachusetts, United States.
- [2] Murray, J. (2009). *Oxford English Dictionary*, Oxford University Press, Oxford, United Kingdom.
- [3] Eves, H. W. (1969). *An Introduction to the History of Mathematics*, third ed.,

- Holt, Rinehart and Winston, New York, United States.
- [4] Elésèmoyò, I. O. (2018). Design and Implementation of a Numeral-in-Context English-to-Yorùbá Text Translation System. An Unpublished M.Sc. Thesis, *Ọbáfẹmi Awólówọ University, Ilé-Ifẹ, Nigeria*.
- [5] Mann, A. (1887). Notes on the Numeral System of the Yorùbá Nation, *The Journal of the Anthropological Institute of Great Britain and Ireland*. 16 (1887):59–64.
- [6] Conant, L. L. (1896). *The Number Concept: Its Origin and Development*, MacMillan and Company, New York, United States, 1896.
- [7] Dennett, R. (1917). How the Yorùbá count: And the universal order in creation, etc., *Journal of the Royal African Society*. 16 (1917):242–250.
- [8] Dennett, R. (1917). How the Yorùbá count (continued), *Journal of the Royal African Society*. 17 (1917):60–71.
- [9] Hall, G. (1973). Yorùbá Numerals, Folklore Forum Bibliographic and Special Series, No. 11, *Studies in Yorùbá Folklore*. (1973):1–6.
- [10] Ekúndayò, S. A. (1977). Vigesimal Numeral Derivational Morphology: Yorùbá Grammatical Competence Epitomized, *Anthropological Linguistics*. 19 (1977): 436–453.
- [11] Awóbùlúyì, O. (2016). *Ẹkọ̀ Ìsèdà-Ọ̀rọ̀ Yorùbá*, Kingdom Arts Publishing, Ibadan, Nigeria.
- [12] Lóngé, O. (2009). *A Yorùbá Decimal Number System*, Bookbuilders, Ibadan, Nigeria.
- [13] Lóngé, O. (2009). *A Yorùbá Vigesimal Number System*, Bookbuilders, Ibadan, Nigeria.
- [14] Akinadé, O. O. and Ọdẹjọbí, O. A. (2014). Computational Modelling of Yorùbá Numerals in a Number-to-Text Conversion System, *Journal of Language Modelling*. 2 (2014):167–211.
- [15] Elesemoyo, I. O. and Odejobi, O. A. (2022). Machine translation system for numeral in English text to Yorùbá language, *Ife Journal of Information and Communication Technology*. 6 (2022):26–37.
- [16] Jimoh, E., Akinlolu, A., Akanbi, M. and Olojede, O. (2023). An android-based Yorùbá numeral translation system, *FUOYE Journal of Pure and Applied Sciences. (FJPAS)*. 8(3):38–46.
- [17] Elesemoyo, I. O. and Odejobi, O. A. (2022). Yorùbá counting system versus roman, Chinese, Hindu-Arabic, Hausa and Igbo counting systems: A computational comparison, in: M. E. Ekpenyong, I. I. Udoh (Eds.), *Current Issues in Descriptive Linguistics and Digital Humanities: A Festschrift in Honor of Professor Eno-Abasi Essien Urua*, Springer Nature Singapore, Singapore, pp. 645–654. URL: https://doi.org/10.1007/978-981-19-2932-8_43.
- [18] Agbeyangi, A. O., Eludiora, S. and Popoola, O. (2016). Web-based Yorùbá numeral translation system, *International Journal of Artificial Intelligence*. 5(2016):127–134.
- [19] Ògúnbòwálé, P. O. (1967). *The Essentials of the Yorùbá Language*, Hodder and Stoughton, London, United Kingdom, 1967.
- [20] Johnson, S. (1921). *The History of the Yorùbás: From the earliest times to the beginning of the British Protectorate*, Cambridge University Press, Lagos, Nigeria, 1921.
- [21] Abraham, R. C. (1962). *Dictionary of Modern Yorùbá*, University of London Press, London, United Kingdom.
- [22] Archangeli, D. and Pulleyblank, D. (1989). Yorùbá vowel harmony. *Linguistic inquiry*. 20(2): 173–217.
- [23] Akinlabí A. (2004). *The Sound System of Yorùbá*, pp.453–468, Africa World Press, Trenton, NJ08607

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